

MAXWELL'S SPHERE AS THE χ -CELL: Deriving c as a Geometric Phase Lock and $K = 1/\pi$ (v.02).

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ABSTRACT

The synchronization coefficient $K = 1/\pi$, previously introduced as a postulate in the Independent Convergence paper, is here shown to follow without postulation from the ratio of two geometrically distinct closure conditions on the Maxwell spherical resonator. Furthermore, c is redefined not as a kinematic propagation speed, but as the unique *impedance-invariance velocity* for which the vacuum impedance $Z_0 = \sqrt{\mu_0/\epsilon_0}$ remains constant over a full phase cycle. Spin-1/2 and the topological relation $\pi_1(SO(3)) = \mathbb{Z}_2$ follow immediately as a consequence of the geometric half-cycle structure of the $E \leftrightarrow H$ transfer within the same sphere.

1. The starting point: Maxwell's sphere is not empty

Maxwell derived c from the geometry of a closed spherical cavity. Inside that cavity, boundary conditions require the tangential component of E to vanish on the surface—the fields E and H cannot both be zero inside. The sphere is, by construction, a resonator with a minimum non-zero excitation. This is the physical content of the statement $E \neq 0$ always: a closed spherical boundary geometrically forbids a zero-field interior. The vacuum is not empty- it is the minimum mode of Maxwell's resonator.

We identify this sphere with the phase cell (χ -cell), characterised by the two foundational relations:

$$\tau = \hbar/E \text{ and } s = \hbar c/E = c \cdot \tau$$

where τ is the duration of one radian of phase and s is the spatial radius of the cell. These two relations are definitions—they reproduce Minkowski spacetime exactly, and $c = s/\tau$ is an identity by construction. The non-trivial content arises when two geometrically distinct closure conditions are imposed on the same sphere.

2. Two closure conditions on the same sphere

2.1 Phase closure condition

The phase cycle closes when the accumulated phase over the diameter equals one radian—the minimum quantum of phase in the framework: $k \cdot s = 1$ This is the condition that the τ -field completes one cycle internally without discontinuity.

2.2 Electromagnetic closure condition (Maxwell)

The first standing-wave mode of a spherical electromagnetic resonator requires the wavevector k to satisfy: $k \cdot s = \pi$ This is the condition for the lowest-order TE or TM mode: the field completes half an electromagnetic oscillation across the diameter, with E vanishing on the surface.

2.3 The ratio: $K = 1/\pi$ derived

The ratio of these two conditions, evaluated at the same radius s for the same sphere, is: $K = (k \cdot s)_{\text{phase}} / (k \cdot s)_{\text{Maxwell}} = 1/\pi$ $K = 1/\pi$ is not a postulate. It is the geometric ratio between the phase condition and the electromagnetic resonance condition of the Maxwell sphere. This closes the open item in the Independent Convergence paper, where $K = 1/\pi$ was introduced without derivation.

3. Why c is fixed: Z_0 Invariance and the Topological Phase Lock

The interior of the Maxwell sphere contains both E and H fields. Their ratio defines the vacuum impedance:

$$Z_0 = E/H = \sqrt{\mu_0/\epsilon_0} \approx 376.73 \Omega$$

The phase cycle τ requires that this ratio remains absolutely constant over the full phase cycle - that is, Z_0 must be invariant as the phase completes one radian and the energy transfers between E and H .

The condition for Z_0 -invariance across the cycle uniquely fixes the topological sweep rate:

$$c = 1/\sqrt{\epsilon_0\mu_0}$$

Visually, c acts as a geometric "phase lock". Any other sweep rate would cause Z_0 to shear within the cycle—the E and H vectors would fall out of synchronization, tearing the internal standing wave and violating the closure condition. Therefore, c is not a kinematic velocity of an object moving through space; it is the unique geometric invariant that holds the Maxwell sphere intact. Maxwell measured ϵ_0 and μ_0 independently and computed c from their product—not knowing he was computing the Z_0 -invariance condition of the χ -cell.

4. Spin-1/2 as a Half-Cycle Consequence of Rotation Geometry

The electromagnetic closure condition $k \cdot s = \pi$ means that in one phase period τ , the EM field inside the sphere completes exactly π radians—a half-oscillation, transferring energy from E to H . A full $E \rightarrow H \rightarrow E$ physical cycle requires 2τ , corresponding to a 2π physical rotation.

However, in the non-Euclidean topology of the χ -cell, returning to the absolute original phase state (a full spinor closure) requires unraveling the internal phase sweep. A full spinor closure requires 4τ , corresponding to 4π . This is precisely the physical content of $\pi_1(SO(3)) = \mathbb{Z}_2$: the fundamental group of rotations dictates that the minimum closed path to untangle the space requires a 4π rotation, not 2π .

Spin-1/2 follows directly from the half-cycle structure of the $E \leftrightarrow H$ transfer within the Maxwell sphere. It is not postulated, but is a geometric consequence of the exact same topology that fixes K and c .

5. Conclusion

From Maxwell's sphere and the two foundational relations ($\tau = \hbar/E$, $s = \hbar c/E$), four results follow without new postulates:

1. $K = 1/\pi$ as the ratio of two geometric closure conditions.
2. c as the topological Z_0 -invariance speed of the χ -cell.
3. Spin-1/2 as the geometric consequence of the 4π spinor closure.
4. $E \neq 0$ always as the minimum resonator boundary condition.

Minkowski spacetime is reproduced in the macroscopic limit. Maxwell's sphere is not a historical analogy—it is the physical foundation from which the entire Phase-Dynamic architecture follows.

7. Declarations

Ethical Approval and Consent to Participate: Not applicable. (This study does not involve human participants or animals).

Consent for Publication: Not applicable. (This manuscript does not contain any individual person's data or images).

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Declaration on the Use of Artificial Intelligence and Large Language Models:

During the preparation of this manuscript, the author used the Gemini large language model (developed by Google) strictly for English language editing, formatting, and proofreading to ensure grammatical clarity. The AI tool was not used for conceptualization, data generation, or the

creation of scientific ideas. The author explicitly states that all core intellectual content - specifically, fundamental concepts and ideas comprising the "Phase Dynamics" theory - was conceived, developed, and formulated entirely by the sole human author. The author has thoroughly reviewed, verified, and approved the final output, and accepts full scientific and legal responsibility for the integrity, factual accuracy, and originality of the entire manuscript. To ensure absolute mathematical integrity, formal logical consistency, and numerical reproducibility, the entire theoretical and computational framework of this monograph underwent a rigorous, multi-tiered cross-verification protocol utilizing advanced artificial intelligence platforms alongside standard industrial integrated development environments (IDEs). The analytic derivations - including the tensor Maxwell equations in non-Euclidean toroidal coordinates, the Helmholtz wave profiles, and the precise lepton mass matrix equations—were independently cross-checked and verified for logical and algebraic closure across three state-of-the-art large language models: Gemini (Google), DeepSeek, and Claude (Anthropic). This cross-model synthesis eliminated potential analytical oversight, ensuring that the foundational invariants of Phase Dynamics are free from structural anomalies. Simultaneously, all algorithmic solvers, Finite-Difference Time-Domain (FDTD) execution scripts for the Maxwell-Kuramoto-Rabi-Loschmidt simulations, and data-visualization pipelines were developed, debugged, and stress-tested within VS Code and executed via cloud-based Google Colab environments. The numerical outputs confirming the lepton mass predictions and topological phase limits were computationally validated against floating-point precision standards. By implementing this modern, transparent open-science methodology, the author guarantees that all mathematical assertions and quantitative predictions presented herein are fully verifiable, deterministic, and algorithmically reproducible.

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